

Milestone - 3

Milestone 3 (Weeks 5–6) – CRM Integration & Conversation Intelligence

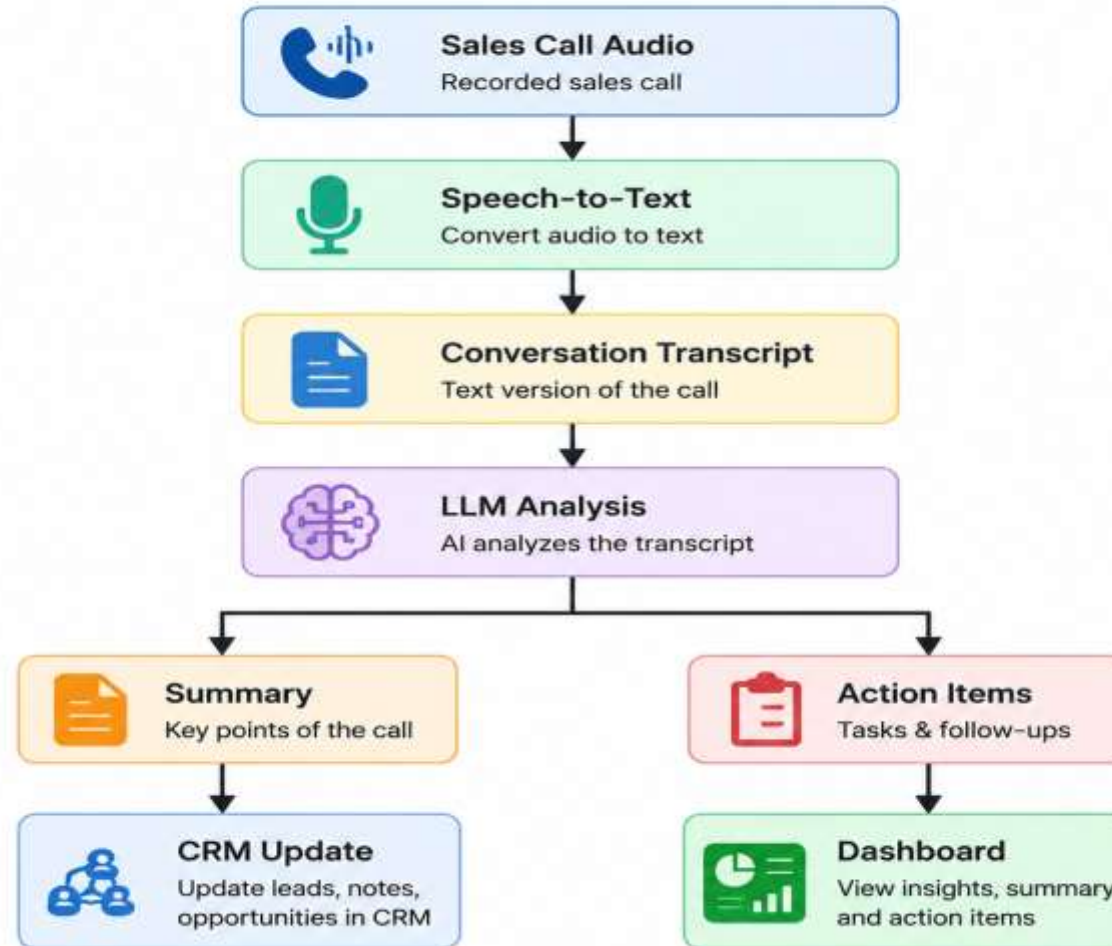
- Integrate CRM systems and sales workflows.
- Implement meeting and sales conversation summarization.
- Develop FastAPI endpoints for external integrations.

Objective

- Build an AI-powered system that:
- Connects with CRM platforms.
- Stores customer and sales information.
- Summarizes sales meetings automatically.
- Extracts insights from conversations.
- Provides FastAPI endpoints for external applications.

Architecture

AI SALES INTELLIGENCE PIPELINE



Part 1: CRM Integration

Step 1: Install Required Packages

```
pip install fastapi uvicorn requests sqlalchemy
```

Step 2: Create CRM Database Model

```
from sqlalchemy import Column, Integer, String
from database import Base
```

```
class Customer(Base):
    __tablename__ = "customers"

    id = Column(Integer, primary_key=True)
    name = Column(String)
    email = Column(String)
    company = Column(String)
    status = Column(String)
```

Explanation

Stores:

- Customer Name
- Email
- Company
- Lead Status

Example:

Name: John

Company: Microsoft

Status: Interested

Step 3: Create CRM Service

```
customers = []
```

```
def add_customer(data):  
    customers.append(data)  
    return data
```


Step 4: CRM API Endpoint

```
from fastapi import FastAPI
```

```
app = FastAPI()
```

```
@app.post("/customer")  
def create_customer(customer: dict):  
    customers.append(customer)  
    return {"message": "Customer Added"}
```

Run:

```
uvicorn main:app --reload
```

Test:

POST /customer

Response:

```
{  
  "message": "Customer Added"  
}
```

Part 2: Conversation Intelligence

- **Step 1: Sample Sales Transcript**

transcript = """

Customer is interested in purchasing
our AI analytics platform.

Requested pricing details.

Follow-up next Tuesday.

"""

Step 2: Generate Summary

Using OpenAI or another LLM:

```
from openai import OpenAI

client = OpenAI()

response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[
        {
            "role": "user",
            "content": f"Summarize:\n{transcript}"
        }
    ]
)

print(response.choices[0].message.content)
```

Output:

Customer interested in *AI* platform.

Requested pricing.

Follow-up scheduled Tuesday.

Step 3: Extract Action Items

Prompt:

```
prompt = f"""
```

Extract action items:

```
{transcript}
```

```
"""
```

Output:

1. Send pricing document.
2. Schedule follow-up Tuesday.

Step 4: Sentiment Analysis

Install:

```
pip install textblob
```

Code:

```
from textblob import TextBlob
```

```
text = "I really like this product."
```

```
analysis = TextBlob(text)
```

```
print(analysis.sentiment)
```

Output:

Polarity = 0.8

Meaning:

Score

>0

<0

0

Meaning

Positive

Negative

Neutral

Part 3: Meeting Summarization Service

- **Create Utility Function**

```
def summarize_meeting(transcript):  
    summary = transcript[:200]  
    return summary
```

API Endpoint

```
@app.post("/summarize")
def summarize(data: dict):

    transcript = data["transcript"]

    summary = summarize_meeting(transcript)

    return {
        "summary": summary
    }
```

Request:

```
{  
  "transcript": "Meeting discussion..."  
}
```

Response:

```
{  
  "summary": "Meeting discussion..."  
}
```

Part 4: Conversation Insights

- Extract:
- Customer Interest
- Budget Mention
- Competitors Mentioned
- Next Meeting Date

Example:

```
insights = {  
  "interest": "High",  
  "budget": "$5000",  
  "next_meeting": "2026-08-01"  
}
```

Endpoint

```
@app.post("/insights")
```

```
def get_insights():
```

```
    return {
```

```
        "interest": "High",
```

```
        "budget": "5000"
```

```
    }
```

Part 5: FastAPI External Integration

Endpoint for CRM Push

```
@app.post("/crm/push")
def push_to_crm(data: dict):

    return {
        "status": "Success",
        "customer": data
    }
```

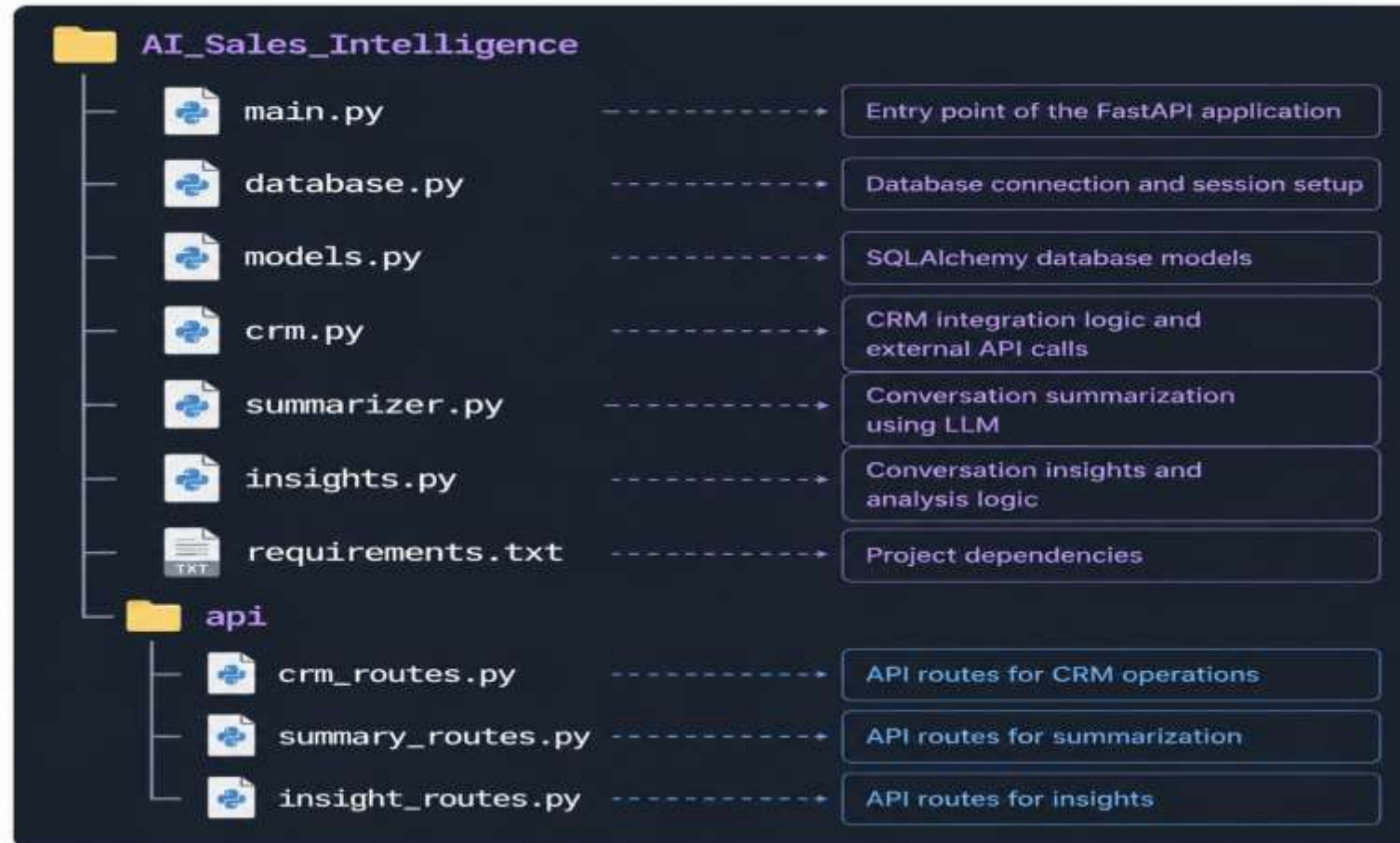
Endpoint for Summary

```
@app.get("/summary/{meeting_id}")
def summary(meeting_id:int):

    return {
        "meeting_id":meeting_id,
        "summary":"Customer interested."
    }
```

Folder Structure

AI_Sales_Intelligence/



CRM Integration & Conversation Intelligence

Sync with CRM and extract insights from sales conversations

Leads

Outreach

Conversations

Dashboard

CRM Sync Status

Synced

Contact Synced

Added

Sarah Johnson, CTO at TechCorp Solutions

Salesforce

2 min ago

Activity Logged

Updated

Initial outreach email sent and opened

HubSpot

15 min ago

Deal Stage Updated

Moved

Moved from "Prospecting" to "Qualified"

Salesforce

1 hour ago

Task Created

New

Meeting Summary

AI Powered

Sarah Johnson, CTO

45 min • Today

TechCorp Solutions

Key Discussion Points

- Data processing bottlenecks affecting customer experience
- Need for real-time analytics and reporting capabilities
- Budget approved for Q3 technology infrastructure upgrade
- Competitive evaluation in progress with 2 other vendors

Action Items

Alex Thompson

Send technical architecture document and integration guide

Due: Jun 10

Sarah Johnson

Schedule technical deep-dive with engineering team

Due: Jun 12

Recent Activity

Follow-up email opened by Sarah

TechCorp Solutions

30 min ago

Discovery call completed (45 min)

TechCorp Solutions

2 hours ago

Demo scheduled for Jun 15 at 2:00 PM

TechCorp Solutions

Yesterday

Added note: "Technical team very interested in API capabilities"

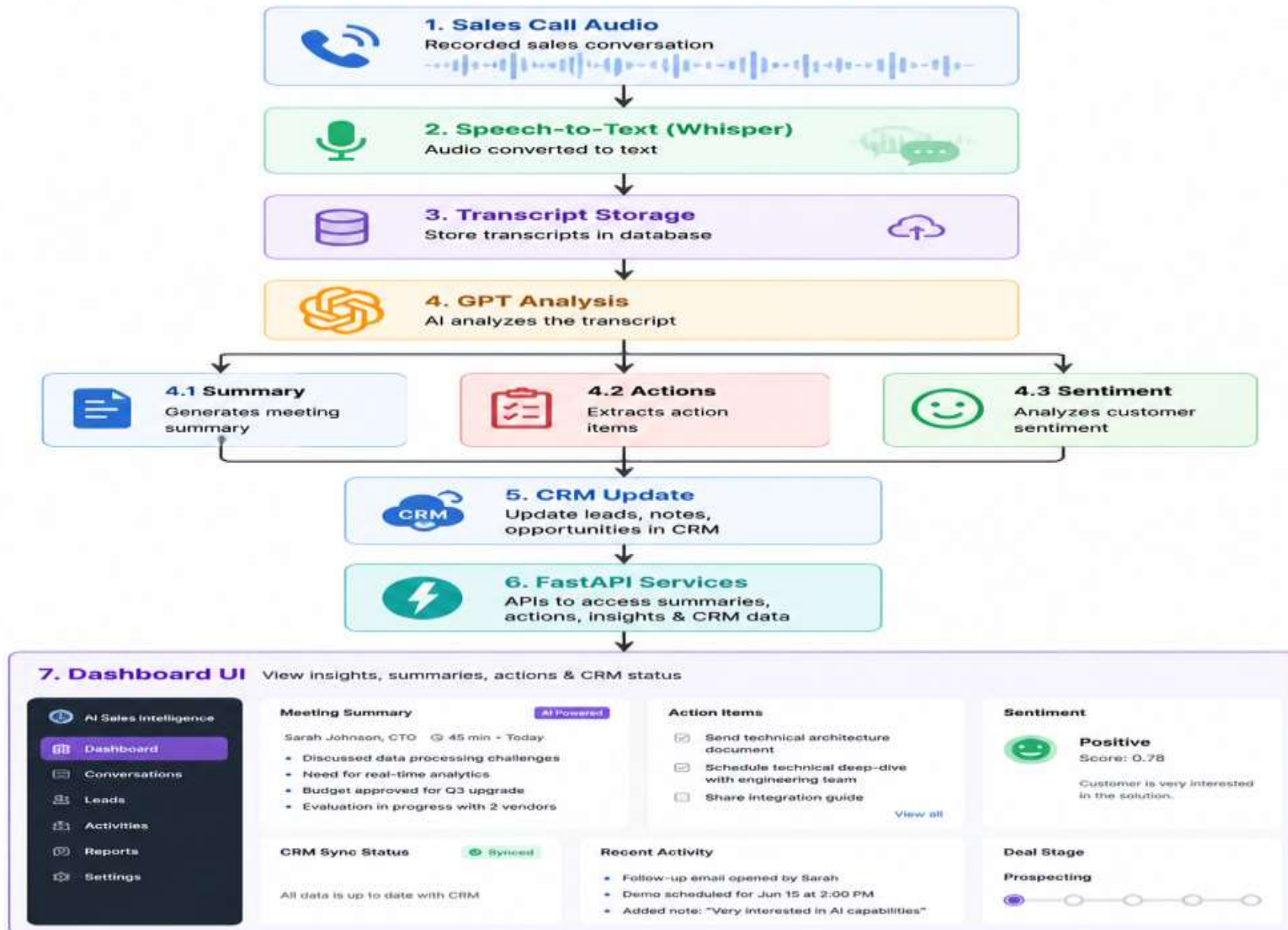
TechCorp Solutions

Yesterday

Initial outreach email sent

TechCorp Solutions

3 days ago



7. Dashboard UI

View insights, summaries, actions & CRM status

AI Sales Intelligence

Dashboard

Conversations

Leads

Activities

Reports

Settings

Meeting Summary

AI Powered

Sarah Johnson, CTO · 45 min · Today

- Discussed data processing challenges
- Need for real-time analytics
- Budget approved for Q3 upgrade
- Evaluation in progress with 2 vendors

CRM Sync Status

Synced

All data is up to date with CRM

Action Items

- ☒ Send technical architecture document
- ☒ Schedule technical deep-dive with engineering team
- ☐ Share integration guide

View all

Recent Activity

- Follow-up email opened by Sarah
- Demo scheduled for Jun 15 at 2:00 PM
- Added note: "Very interested in AI capabilities"

Sentiment

Positive

Score: 0.78

Customer is very interested in the solution.

Deal Stage

Prospecting